Yanqi Feng 冯焱琦

Ph.D. Student | Shanghai Jiao Tong University

School of Automation and Intelligent Sensing · Underwater robot control and planning

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Education

Shanghai Jiao Tong University Sep 2026 - Present

Ph.D. student, School of Automation and Intelligent Sensing | Advisor: Prof. Jianping He

Xi'an Jiaotong University Sep 2022 - Jun 2026

Undergraduate studies in Automation, XJTU Academic Elite Program (Yuejie Class)

Faculty of Electronic and Information Engineering; Qian Xuesen College | Advisor: Prof. Deqiang Han

- Academic score: **98.68**; rank: **1/195** in Automation and **1/30** in the Yuejie Class.
- Selected for the Yuejie talent development program, which admits approximately 35 freshmen annually.
- Foundation courses: Advanced Mathematics I-1/2 (99/94), Linear Algebra (100), University Physics II-1/2 (100/98), and Circuits (100).
- Major courses: Computer Networks (100), Operations Research (99), Automatic Control (97), and Digital Signal Processing (95).

Publications and Preprints

[1] Yuheng Zhou, Haiyang Cheng, **Yanqi Feng**, Pangkit Fong, Mei Xuan Lee, Marcus Gee, Chongrong Fang, Jianping He. **Underwater Visual Target Tracking with Target-Specific Depth Estimation and Adaptive Model-Fusion Predictive Control.** Preprint, arXiv:2609.20731 [cs.RO], 17 September 2026. [arXiv][PDF]

[2] **Yanqi Feng**, Enming Chu, Junru Chen, Long Chen. **A Two-Stage Search Algorithm for Static Target in A 2D Unknown Complex Environment.** 2025 2nd International Seminar on Artificial Intelligence, Computer Technology and Control Engineering (ACTCE). IEEE, 2025, pp. 70-75. [DOI]

Research Experience

EKF with a 9-Axis IMU and a Hydrostatic Pressure Sensor Aug 2025

Independent project, Shanghai Jiao Tong University | Advisor: Prof. Jianping He

- Developed an extended Kalman filter for underwater robot attitude estimation by fusing 9-axis IMU and pressure measurements on the IWIN-FINS platform; implemented the embedded code.
- Experiments showed improved stability compared with an IMU-only Mahony algorithm.

Deadlock Control and Scheduling with Limited Buffers Feb 2024 - Jan 2025

Independent project, Xi'an Jiaotong University | Advisor: Assoc. Prof. Yanxiang Feng

- Studied deadlocks in a two-stage assembly no-wait permutation flowshop with limited buffers; enhanced discrete particle swarm optimization (DPSO) using Q-learning for dynamic evolution strategy selection.
- Proposed a two-stage local search (VND+IG) to escape local optima; the algorithm outperformed HSSA, HGA, and ABC in experiments.

Design and Development of an AI-Powered Oilfield Terminal Apr 2024 - Apr 2025

National Undergraduate Innovation Training Program | Advisor: Cuicui Zhang, XJTU

- The project used convolutional neural networks and optimization to classify oil-well indicator diagrams, estimate operating conditions, fluid level, and production, and support automatic intermittent pumping on an RK3568 embedded device.
- Built the dataset of abnormal oil-well indicator diagrams.

Open-Source Project

FinsROV: An Underwater Camera-Based Multi-Robot Platform Contributor

Open hardware, embedded and host software, and step-by-step video tutorials

- Debugged embedded robot control code and host-controller communication; produced video tutorials.
- Contributed to an open-source platform spanning mechanical design, software, and practical robot-building guidance.

GitHub: IWIN-FINS / FinsROV | Video tutorials

Competition Experience

MATE ROV Competition, China Mainland Regional Apr 2026

Team Captain | Advisor: Prof. Jianping He, Shanghai Jiao Tong University

- Won the **PIONEER division championship** on the team's first appearance.
- Built an underwater robot with a four-degree-of-freedom manipulator for 3D reconstruction, species recognition, target grasping, and precision manipulation.
- Developed YOLO-based underwater species recognition; organized the technical architecture and documentation, coordinated modules, and managed preparation milestones.

Mathematical Contest in Modeling (MCM) Feb 2024

Team Leader / Lead Writer | Advisor: Prof. Long Chen, Xi'an Jiaotong University

- Received the **Finalist Award (top 2%)**.
- Modeled the location of a wrecked submersible in a complex deep-sea environment and proposed a two-stage search strategy incorporating A*.
- Led paper writing, LaTeX typesetting, figure preparation, modeling support, and team coordination; the search-algorithm work led to the ACTCE 2025 publication.

China Robot Competition and RoboCup China Open Nov 2024

Embedded Development | Advisors: Ling Guo and Qiang Chen

- Won **Second Prize, Automatic Sorting Robot - Intelligent Delivery**.
- Implemented information processing, motion control, and behavior logic for an STM32F103ZET6-based vehicle that sorts and delivers mail of different colors.
- Responsible for motion planning, communication control, and hardware testing.

Honors and Awards

National Scholarship, twice 2023-2024; 2024-2025

Yuejie Scholarship: RMB 60,000 base + RMB 180,000 international exchange 2022 - 2026

Outstanding Student Award 2022-2023; 2023-2024

Silver Prize, China International College Students' Innovation Competition, Shaanxi Sep 2024

Second Prize, 15th National College Mathematics Competition, Shaanxi (Non-Math A) Dec 2023

Six university-level awards, including:

- FLTRP ETIC Cup campus contest: Second Prize in reading and Third Prize in writing (Sep 2023).
- Seventh place, one-minute eight-person long-rope event, 61st XJTU Sports Meeting (May 2024).
- First Prize, three-minute ten-person long-rope event, XJTU Spring Rope-Skipping Competition (Apr 2024).

Skills and Languages

Programming C, MATLAB, Python

Tools Keil, Git, Docker, LaTeX

English CET-4: 613; CET-6: 547